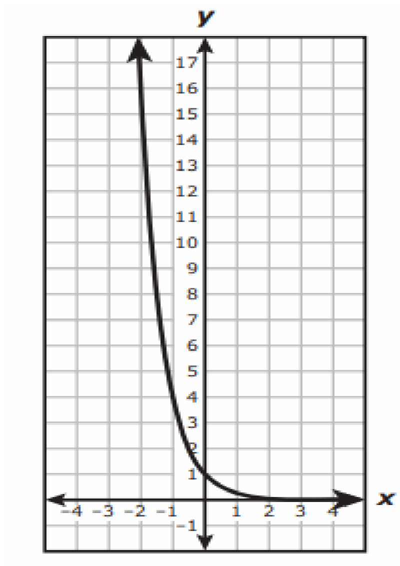
Fourth Six Weeks Exam Review

1) Consider the function .

a) What is the domain? b) What is the range? c) What is the asymptote?

2) An exponential function is graphed on the right.



Which function is represented by the graph?

A) $p(x) = (0.25)^x$

B) $p(x) = 2(0.5)^x$

C) $p(x) = (1.25)^x$

D) $p(x) = (25)^x$

3. a) Feta's weight changed from 2 ounces to 6 ounces. What is the percent change in her weight?

   b) Then Feta went on a special diet, and went back down to 4 pounds. What is this percent change?

4. There were 1500 roaches living in a tree. The population increases at a rate of 12% per week.
   a) Write an equation to model the roach growth.

   b) Use your equation to determine when there will be 10,000 roaches.

5. A roach population in Mr. Jackson's room can be modeled by the equation $R = 1250(0.975)^x$, where x is the amount of time in days since a roach trap was put out.
   a) What is the decay rate of the roach population?

   b) What is the decay factor?

   c) How long will it be before there are only 1000 roaches?

   d) What is the half-life of this roach population?

6. Given the functions $f(x) = 0.7(1.2)^x$ and $g(x) = 1.2(0.7)^x$
   a) Identify which function is exponential growth and which is exponential decay. How do you know?

   b) For the growth function, identify the starting value, the growth factor, and the growth rate.

   c) For the decay function, identify the starting value, the decay factor, and the decay rate.

7. Write an equation that represents the data in each of the given tables.

a)

x	y
1	250
2	125
3	62.5
4	31.25

b)

x	y
-1	250
0	750
1	2250
2	6750

8. How can you determine if (a) a table, (b) a graph, and (c) an equation is exponential?

9. How can you determine if an exponential relationship is growth or decay?

10. WITHOUT USING DESMOS, make a table of values and a high-quality graph of $a(x) = (\frac{1}{2})^x$ with the domain of all integers such that $-3 \le x \le 3$. Show your work!

Fourth Six Weeks Exam Review Answer Key

1a) D: all real numbers 1b) R: $y > 3$ 1c) asymptote: $y = 3$
3a) 200% increase 3b) $33\,^1/_3$ %
4a) $y = 1500(1.12)^x$ 4b) between 16 and 17 weeks
5a) 2.5% 5b) 0.975 5c) 9 days 5d) between 27 and 28 days
6a) first equation is growth, because the growth factor is greater than 1;
second equation is decay, because it's multiplying factor is less than 1
6b) starting value = 0.7; growth factor = 1.2; growth rate = 20%
6c) starting value = 1.2; decay factor = 0.7; decay rate = 30%
7a) $y = 500(\frac{1}{2})^x$ 7b) $y = 750(3)^x$
8a) As x increases in a consistent pattern (usually +1), y is always multiplied by the same constant ratio.
8b) Exponential graphs are curves that are boomerang shaped – they've got one bend in the middle.
8c) Exponential functions have the independent variable (usually x) in the exponent.
9) If the b-value/multiplier/base of the exponent is more than 1, then it's growth.
If it's between 0 and 1, then it's decay.
10)

x	$a(x) = (\frac{1}{2})^x$	y
-3	$(\frac{1}{2})^{-3} = 2^3$	8
-2	$(\frac{1}{2})^{-2} = 2^2$	4
-1	$(\frac{1}{2})^{-1} = 2^1$	2
0	$(\frac{1}{2})^0$	1
1	$(\frac{1}{2})^1$	½
2	$(\frac{1}{2})^2$	¼
3	$(\frac{1}{2})^3$	⅛